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Faculty of Information Technology

Department: Software Engineering

PROJECT FOR FINAL EXAM

Course name: BEST PROGRAMMING PRACTICES AND DESIGN PATTERNS

Instructor: RUTARINDWA JEAN PIERRE
Instructions:

- *This Project will be individually done by every student*
- *Every student must clearly choose the Topic and the case study.*
- *A documentation will be submitted alongside with the software application created.*
- *Marks will be provided after a thorough checking and analysis of what students have done and originality of the work will be checked.*
- *There should be a well-structured or logical connection between phases of the project.*

PHASE 1.

Choose the Topic (**Example:** Online Library Management system) and the case study (**Example:** New Heaven University-NHU) for the software to be developed. Make a thorough System Analysis and design.

Provide the following:

i. Briefly provide the general description and analysis of the chosen case study

ii. Functional Diagram indicating the internal working of the case study (Company chosen)

iii. Identify clearly the problem faced by the company (case study chosen)

iv. Use Object Oriented system analysis and design method and provide the following diagrams:

1. Use case Diagram
2. Class Diagram
3. Activity Diagram
4. Sequence Diagram
5. Component Diagram

PHASE 2.

A software development prototype is an early, incomplete, or experimental version of a software application used to test concepts, validate assumptions, and gather user feedback. It helps to refine functionality, identify design flaws early, and reduce development risks before full-scale coding.

Question:

- Provide a well-summarized software development prototype for the proposed software to be developed. Please, follow software programming best practices appropriate for the software under development. (Please refer from google coding standards specifically for your programming languages you have chosen).
- Select any software Design pattern and clearly show how you have referred from it during the design and development of your software application.

PHASE 3.

- Docker is a tool for running application in isolated environment. Docker is a platform and a tool used for containerization, which is a method of packaging, distributing, and running applications and their dependencies in a consistent environment. Dockerizing a software application involves packaging your app and all its dependencies (libraries, runtime, OS packages) into a single, portable unit called a **container**. It isolates software from its surroundings, ensuring that it will run consistently regardless of the environment in which it is deployed.

Question: Briefly discuss the process to dockerize an application. Then Dockerize your software application developed.

- A Version Control System (VCS) is a tool that tracks and manages changes to software code, allowing multiple developers to collaborate, track file revisions, and revert to previous states. It acts as a secure repository, preventing code loss and resolving conflicts when multiple users work on the same project.

Question: Install and configure any Version Control Systems SVN of your choice. Set the environment so that this version Control system may capture all the necessary parts of your application under development.

PHASE 4.

A software test plan is a detailed document that serves as a roadmap for all testing activities in a project. It outlines what will be tested, how it will be tested, who will do the testing, and the schedule for these activities.

Question: Clearly prepare a software test plan and show how well the software application you have developed will be tested.